

REPUBLIC OF SOUTH AFRICA

PATENTS ACT, 1978

APPLICATION FOR A PATENT AND ACKNOWLEDGEMENT OF RECEIPT

[Section 30 (1)-Regulation 22]

The granting of a patent is hereby requested by the undermentioned applicant on the basis of the present application.

Official Application No.		Applicant's or Agent's Reference
21	01	6848LPS
2026/05164		

71	Full Name(s) of Applicant(s)
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India	

54	Title of invention
A DISTRIBUTED VEHICLE-MOUNTED VISION SYSTEM FOR AUTOMATED TRAFFIC VIOLATION DETECTION AND ELECTRONIC CITATION ISSUANCE	

The applicant claims priority as set out on the accompanying Form P.2. The earliest priority claimed is		
COUNTRY:	NUMBER:	DATE:

This application is for a patent of addition to patent application No.		
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21	01	
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This application is a fresh application in terms of section 37 and based on Application No.		
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21	01	
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This application is accompanied by:		
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X	1.	A single copy of a complete specification of 10 pages.
X	2.	Drawings of 2 sheet(s).
X	3.	Publication particulars and abstract(Form P8)
X	4.	A copy of a figure of the drawing (if any) for the abstract
X	5.	Assignment of invention
	6.	Certified priority document(s)
	7.	Translation(s) of the priority document(s)
	8.	Assignment of priority rights
	9.	A copy of the Form P.2 and the specification of S.A Patent Application (if applicable).
X	10.	A declaration and power of attorney on Form P3
X	11.	Statement on the use of indigenous Biological Resource, Genetic Resource, Traditional Knowledge or Use on Form P26

74	Address of Service:
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA	

Dated this 12th day of May 2026

Digitally signed by : Leon Pierre Susan

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Signature of Applicant(s)

This is returned to the applicant's
address for service as proof of lodging.

RECEIVED
Official Date Stamp
..... Registrar of Patents

REPUBLIC OF SOUTH AFRICA

REGISTER OF PATENTS

PATENTS ACT, 1978

Official application No.		Lodging date: Provisional		Acceptance date	
21	01	2026/05164		22	
International classification		Lodging date: Complete		Granted date	
51	G08G	23	2026/05/12		
71 Full name(s) of applicant(s)/Patentee(s):					
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India					
71 Applicant substituted:				Date registered	
71 Assignee(s):				Date registered	
72 Full name(s) of inventor(s):					
JAHANGIR, Umar Tabrez KHONDRE, Shruti Pradeep KALME, Rushabh Angad JALGAONKAR, Sahil Sanjay DEDGAONKAR, Suruchi G. SHELKE, Priya					
Priority claimed:		Country	Number	Date	
54 Title of invention					
A DISTRIBUTED VEHICLE-MOUNTED VISION SYSTEM FOR AUTOMATED TRAFFIC VIOLATION DETECTION AND ELECTRONIC CITATION ISSUANCE					
Address of applicant(s)/patentee(s):					
666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037 INDIA					
74 Address for service					
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA Reference No. 6848LPS					
61 Patent of addition No.			Date of any change		
Fresh application based on.			Date of any change		

REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
COMPLETE SPECIFICATION
[Section 30(1) – Regulation 28]

OFFICIAL APPLICATION NO.

21	01	2026/05164
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LODGING DATE

22	2026/05/12
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INTERNATIONAL CLASSIFICATION

51	G08G
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FULL NAME(S) OF APPLICANT(S)

71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India
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FULL NAME(S) OF INVENTORS(S)

72	1. JAHANGIR, Umar Tabrez 2. KHONDRE, Shruti Pradeep 3. KALME, Rushabh Angad 4. JALGAONKAR, Sahil Sanjay 5. DEDGAONKAR, Suruchi G. 6. SHELKE, Priya
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TITLE OF INVENTION

54	A DISTRIBUTED VEHICLE-MOUNTED VISION SYSTEM FOR AUTOMATED TRAFFIC VIOLATION DETECTION AND ELECTRONIC CITATION ISSUANCE
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REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
PUBLICATION PARTICULARS AND ABSTRACT
[Section 32(3)(a) – Regulation 2291(g) AND 31]

OFFICIAL APPLICATION NO.		LODGING DATE	ACCEPTANCE DATE
21	01	22	47
2026/05164		2026/05/12	

INTERNATIONAL CLASSIFICATION	NOT FOR PUBLICATION
51	
G08G	CLASSIFIED BY: Wolmarans and Susan Inc.

FULL NAME(S) OF APPLICANT(S)	
71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India

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72	1. JAHANGIR, Umar Tabrez 2. KHONDRE, Shruti Pradeep 3. KALME, Rushabh Angad 4. JALGAONKAR, Sahil Sanjay 5. DEDGAONKAR, Suruchi G. 6. SHELKE, Priya

EARLIEST PRIORITY CLAIMED		DATE
COUNTRY	NUMBER	
33	31	32

TITLE OF INVENTION	
54	A DISTRIBUTED VEHICLE-MOUNTED VISION SYSTEM FOR AUTOMATED TRAFFIC VIOLATION DETECTION AND ELECTRONIC CITATION ISSUANCE

57	The present invention relates to a distributed system configured to be placed in vehicles to automatically detect and report traffic violations, enabling real-time monitoring of roadway environments using mobile sensing units installed in civilian vehicles. The system includes a vision sensing module, an edge embedded processing unit, a geolocation module, a wireless communication interface, and a tamper detection mechanism installed in civilian vehicles, wherein the embedded processing unit analyzes images captured on road surfaces using computer vision and machine learning algorithms to automatically detect traffic rule violations. Upon detection of a violation, a digital record is created including images, identification of the violating vehicle, timestamp, and geolocation coordinates, and this record is transmitted to a central authority via a cellular network for verification and issuance of a digital Challan (traffic ticket). The invention further comprises a tamper detection process designed to track system integrity, generate real-time notifications in case of unauthorized system tampering, and ensure the reliability of gathered evidence, and by deploying a distributed mobile sensing system the invention enables scalable, affordable, and infrastructure-agnostic traffic law enforcement.
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